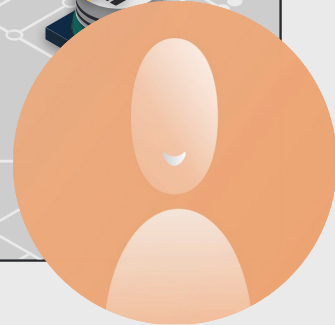
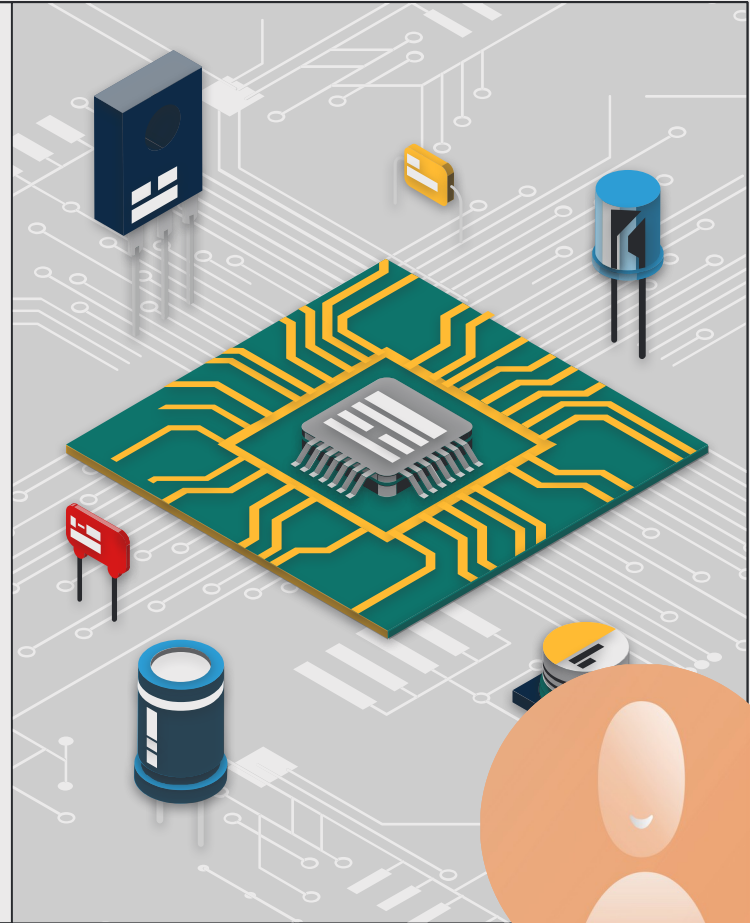


Apple's Just-In-Time

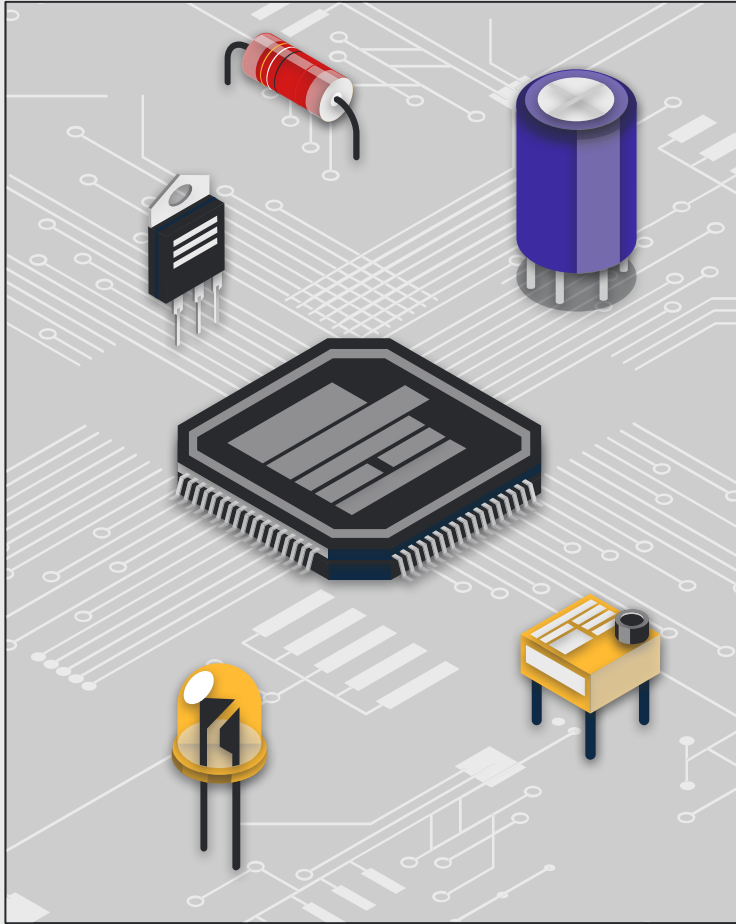
Jeruson Chinnathambi, Laura Smith, Marília
Ackel, Lauren Pounds, Christerpher F. Coleman



Apple Inc.

Multitrillion dollar company.





Disruption in Supply Chain

COVID-19.

What Happened during Covid?

17.9%

Online Work Grew

From 5.7% to 17.9% within
U.S. (U.S. Census Bureau,
2022)

1.6

Students

Shifted Online

1.6 billion students in 188
countries (UNICEF, 2022).



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Demand 

While simultaneously...



Manufacturing

Slowed due to shutdowns. Operated at closed or minimal staff.

(Forbes, Wishart-Smith, 2024)



Single Supplier

Reliance on Taiwan Semiconductor Manufacturing Company.



Assembly

Geopolitical tensions between Taiwan and China affect Apple production.

(Miller, 2025)

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Supply

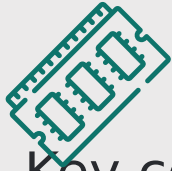
A large, dark blue downward-pointing arrow.

Our Analytical Approach

Diversifying supply chain dependencies and integrating data analytics can support production scheduling, reduce lead-time volatility, and improve demand forecasting—ultimately improving production flow and profitability.

<u>Data Collection</u>	<u>Analytical Tools</u>
Financial reports	Excel
Corporate sustainability & progress reports	Power BI
Policy Analyses	Tableau
Geopolitical risk assessments	Structured Query Language
Labor/ESG data	

Data Collection Process



Key components impacting Operation & Logistics

- Demand of Products – Supply Chain Metrics
- Price Impact on Production – Financials
- Changes in Manufacturing – Suppliers

Data Protection for Users



January to December 2020

- Over 8,600 requests for data
- 82% of requests received data
- 92% of the requests for data were emergencies

Apple's Market Health



27% Current Market
Share

Smartphone Sector



\$269.99 per share

Started at \$22 per share in 1980



2020 Market Share

21% of the smartphone sector



22,429 shareholders

As of the 2025 reported fiscal year

Apple's Economic Health



Increase in R&D and SG&A - 2020

- \$38,668 million in operating expenses

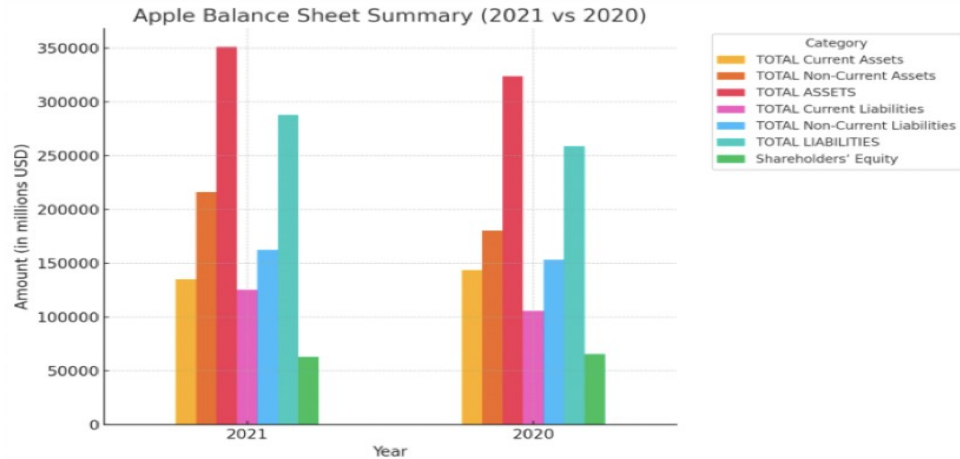


Increase in R&D - 2025

- \$62,151 million in operating expenses. 15% of total net sales due to infrastructure costs

Apple balance sheet

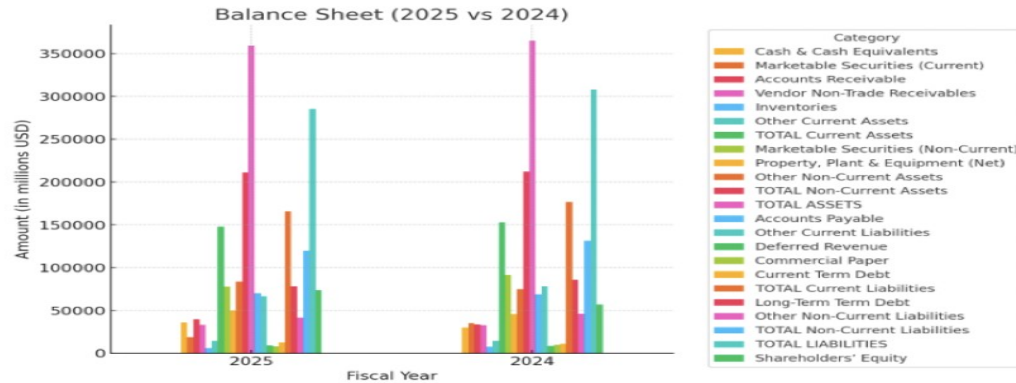
Apple Balance Sheet 2021-2022



Apple's assets and liabilities both increased from 2020 to 2021. This shows strong financial growth during the COVID period as people bought more devices for work and school.

Apple balance sheet

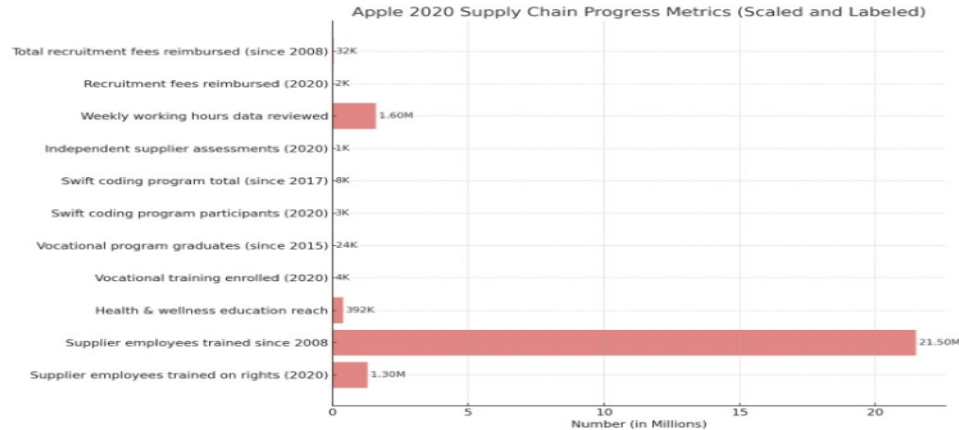
Apple Balance Sheet 2024-2025



Apple showed steady financial performance in 2024 and 2025. A small increase in assets and equity shows stability, while liabilities rose mainly due to long-term debt.

Apple balance supply chain

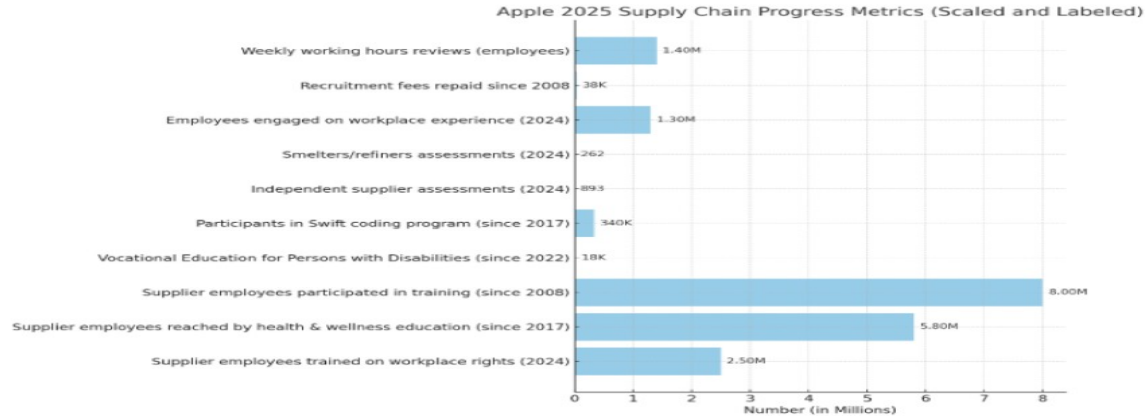
Apple Supply Chain 2020



In 2020, even during COVID, Apple trained over 21 million employees and reviewed 1.6 million working-hour records. This shows Apple kept improving worker conditions during the pandemic

Apple balance supply chain

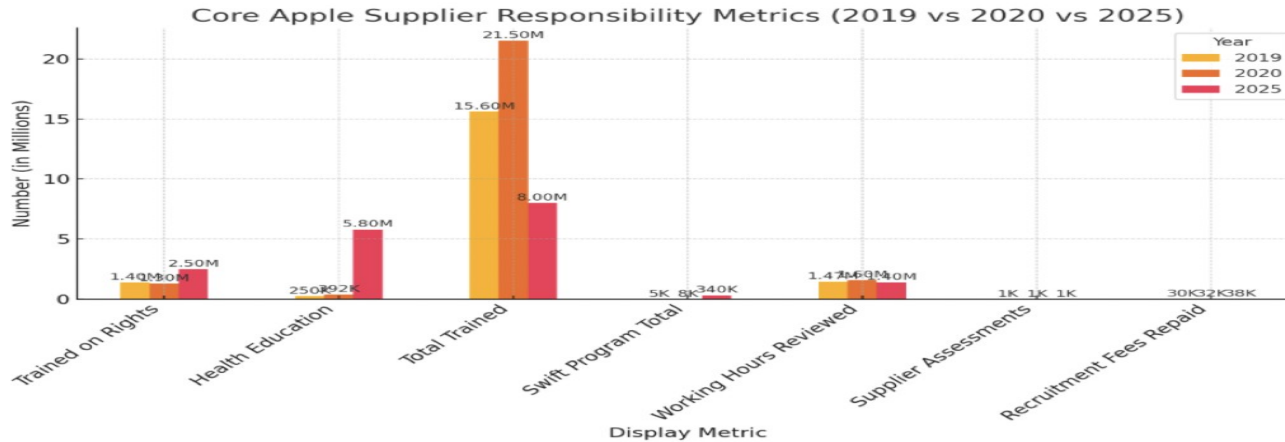
Apple Supply Chain 2025



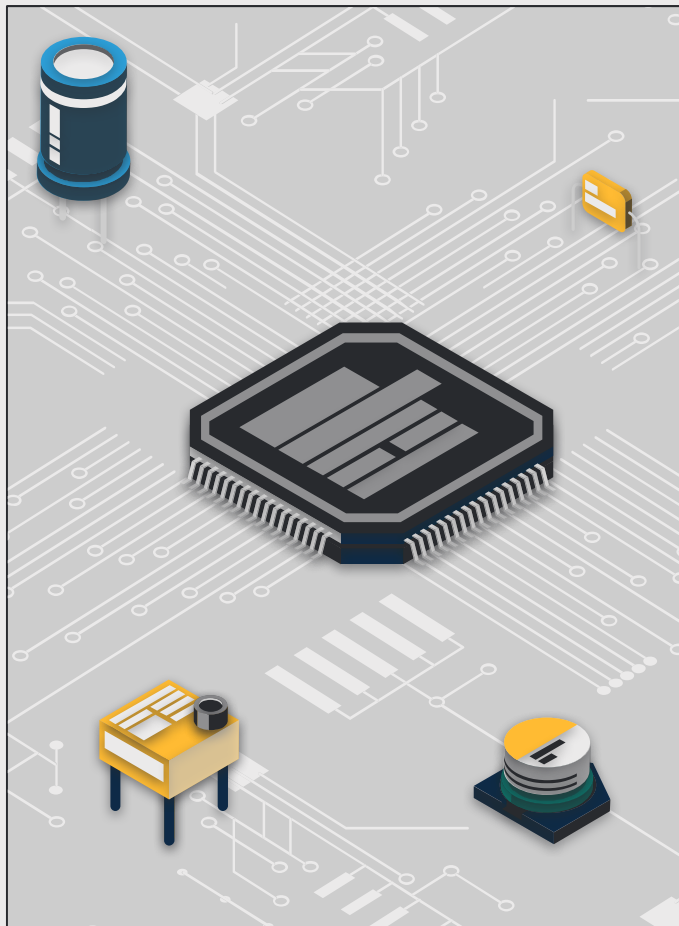
In 2025, Apple expanded training, wellness education, and supplier assessments. Millions of workers received support, showing Apple's continued focus on safety and ethical practices.

Apple balance supplier responsibility

Core Apple supplier responsibility 2019 vs 2020 vs 2025



Apple improved worker training, rights education, and health programs over the years. Total training peaked in 2020 and continued strong in 2025, showing long-term commitment to supplier responsibility



“A prudent man
foreseeth the evil,
and hideth himself
but the simple pass
on, and are
punished.”

—Proverbs 22:3 (KJV)

APPLE MAPS

NORTH AMERICA:
3%

EUROPE: 4-
5%

ASIA:
90%

FACTS

- More than **90 percent** of Apple's supplier facilities are in Asia.
- **Single point of failure** in a global supply chain **worth trillions.**
- Diversification directly addresses the core **structural weakness** in Apple's operating model.

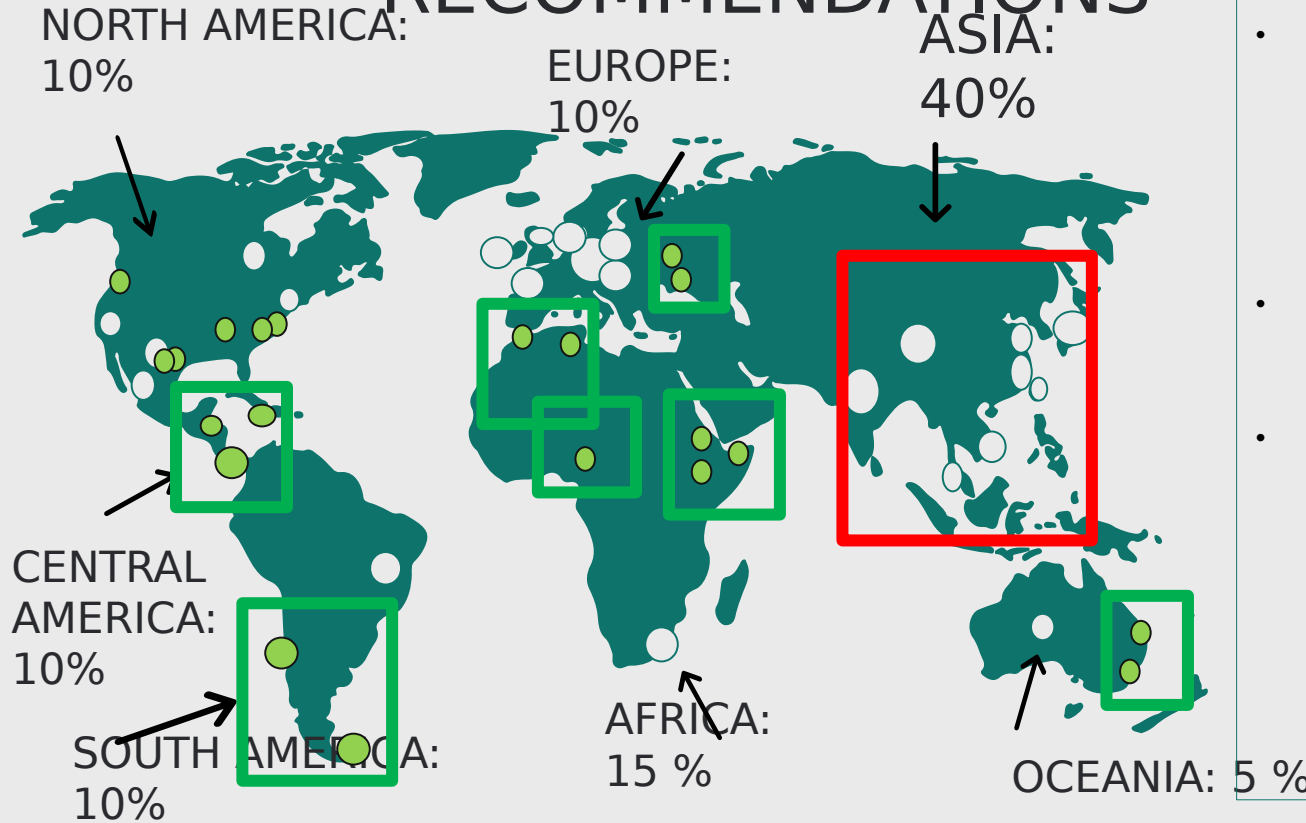
LATIN AMERICA:
1%

AFRICA: >
1%

OCEANIA: >
1%

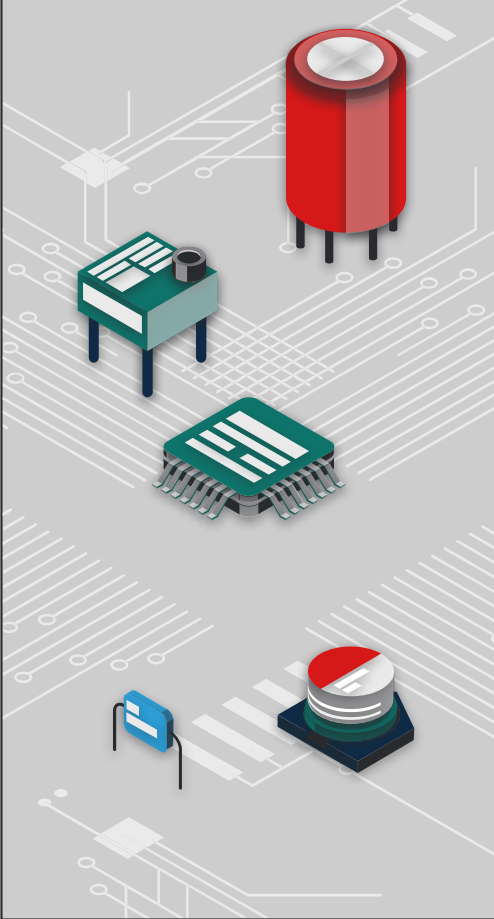


RECOMMENDATIONS



FACTS

- **Reducing Asia to 40%** creates meaningful diversification while preserving the operational advantages of the region.
- **Africa** becomes a **strategic secondary hub**.
- **Central America** becomes a **strategic mid-point region** due to proximity to the United States and access to both Atlantic and Pacific trade routes. The **Panama Canal** offers unmatched routing flexibility.

An isometric illustration of various electronic components on a grey circuit board. The components include a red cylindrical capacitor, a green integrated circuit (chip), a blue and white component, and a stack of three capacitors (red, white, and blue). The circuit board is marked with white lines and circles representing traces and pads.

Beyond Compliance Apple's Key Efforts

- Annual third-party audits
- Publishes full supplier list
- 98% compliance with 60-hour work-week
- 95% of violations corrected within 90 days
- Cut ties with suppliers linked to forced labor

“Look! The wages you failed to pay the workers who mowed your fields are crying out against you. The cries of the harvesters have reached the ears of the Lord Almighty.” James 5:4 (NIV)

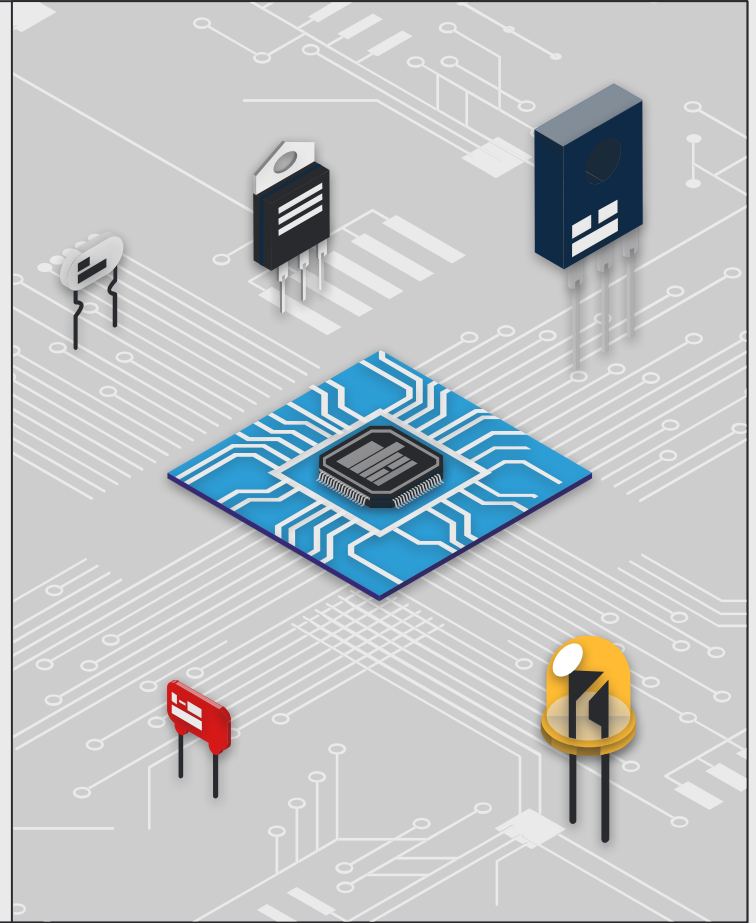
Apple's Supply Chain Persistent Problems

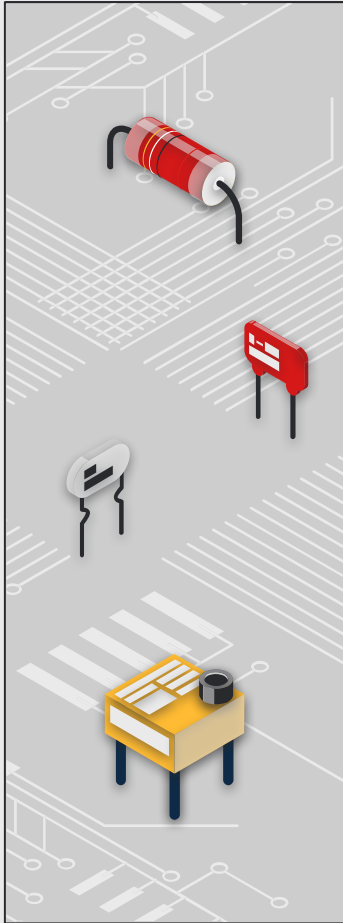
- **Child labor and deaths** in artisanal cobalt mines
- workers doing 18-hour shifts and sleeping in corridors
- German court ruled Apple Watch is not “carbon neutral”
- Excessive use of dispatch workers far above legal limits
- Deep-tier suppliers remain largely outside full oversight (Tier 3/4)



***“Truly I tell you,
whatever you did for one
of the least of these
brothers and sisters of
mine, you did for me.”
Matthew 25:40 (NIV)***

Questions ?





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